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Conflict-Aware User Scheduling and Hybrid Beamforming for Near-Field STAR-RIS Systems

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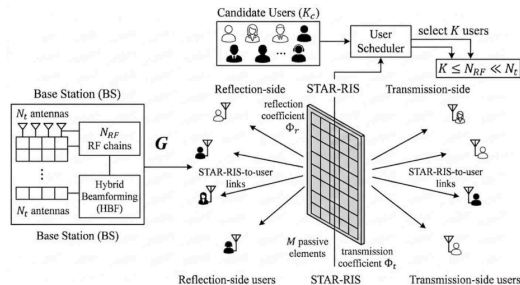
1. Context & Motivation

STAR-RIS: A simultaneously transmitting and reflecting surface that covers both sides of space. Users only receive; the transmission or reflection side is decided by location. The STAR-RIS coefficients are preconfigured for the slot.

Hybrid beamforming: The base station uses a limited number of radio-frequency chains. Each radio-frequency chain supports one analog beam and one scheduled user, followed by a digital precoder.

Constraint: Gain-based scheduling packs users onto the same analog codeword (hurts rate + fairness). Classical SUS is channel-orthogonal but codebook-blind; in the near field, users can share dominant polar codeword b_k^* even when channels are weakly correlated.

2. Overall Architecture



STAR-RIS architecture

Hybrid stream budget. $K \leq N_{RF}$ (one analog beam per RF chain). Φ_T, Φ_R preconfigured; BS outputs S, F_{RF}, F_{BB} .

Near field. Rayleigh $d_{Ray} = 2D^2/\lambda$; paper $d_{Ray}^{BS} \approx 21.26$ m, $d_{Ray}^{RIS} \approx 10.30$ m; BS-RIS / RIS-user / BS-user links inside $\Rightarrow$ polar-domain (angle+range) codebook.

3. Three Factors Aligned in η_k

Proposed CA-SUS-HBF — three ingredients, one screening score

$$\eta_k = \alpha_g \bar{g}_k - \alpha_c c_k + \alpha_o \xi_k$$

gain rewards strength; $-c_k$ penalizes beam fights; $+\xi_k$ favors separable users

(1) Effective gain

$$\bar{g}_k$$

Link strength after
STAR-RIS (how useful
the channel is).

(2) Conflict score

c_k , one number from $\mathcal{C}_k$

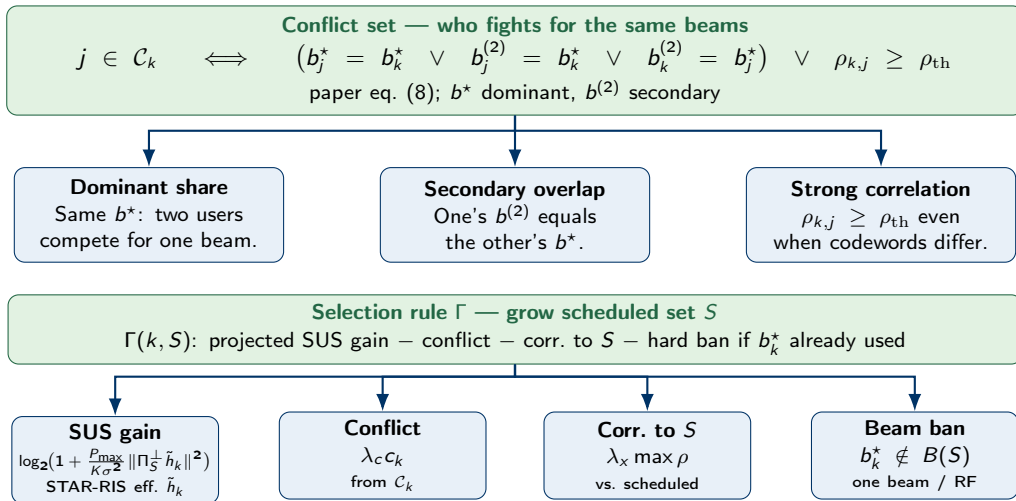
Same favorite, runner-up
match, $\rho_{k,j}$, and own
gap a_k . Not a beam.

(3) Orthogonality

$$\xi_k$$

Channels separable
enough for clean
digital multiuser.

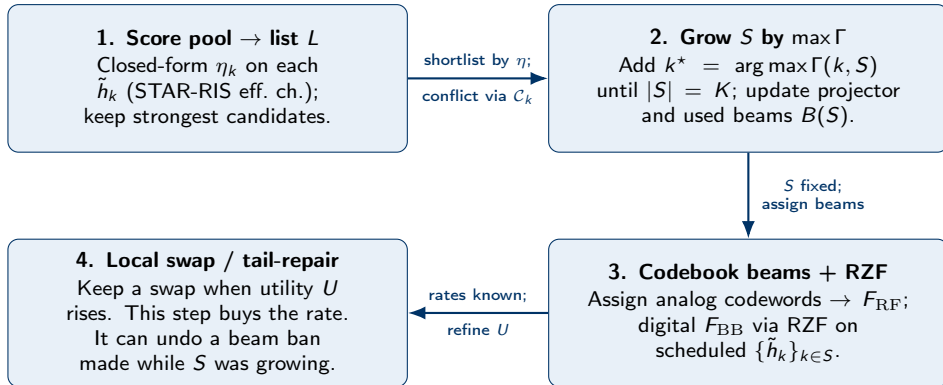
4. Who Conflicts ($\mathcal{C}_k$) and How S Grows (Γ)



Classical SUS is codebook-blind; CA-SUS couples codebook conflict with channel orthogonality.

5. CA-SUS-HBF Pipeline

I/O. Input: candidates, $\{\tilde{h}_k\}$ (STAR-RIS effective channel: direct + cascaded path with Φ inside), polar codebook, budget K . Output: S , F_{RF} , F_{BB} .



6. Simulation Results (Rate)

At 10 dB vs. best practical. CA-SUS-HBF: sum-rate +29.5%, 5th-percentile +76.1%, Jain +0.123.
FD-NF-RZF = fully-digital reference only.

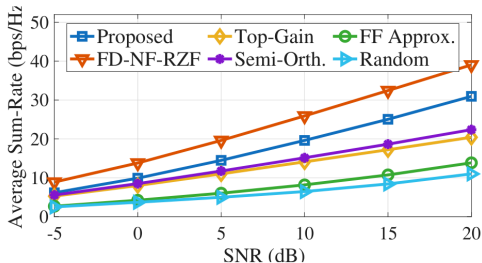


Fig. 1. Average sum-rate versus SNR for different hybrid beamforming schemes.

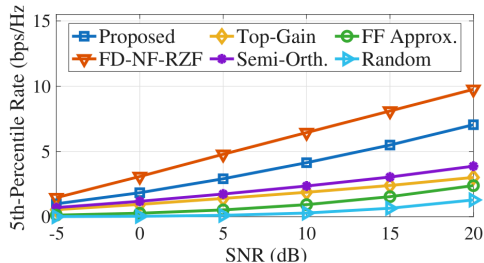


Fig. 2. 5th-percentile scheduled-user rate versus SNR.

7. Simulation Results (Users and Rate–Fairness)

Takeaways of the submitted scheduler. (1) The headline gains are against top-gain and semi-orthogonal scheduling, not the contribution of c_k . (2) Conflict screening matters while the short list $|L|$ is small; at $|L| = 12$ its gain is 0%. (3) The α -sweep still dominates those two baselines in rate and fairness.

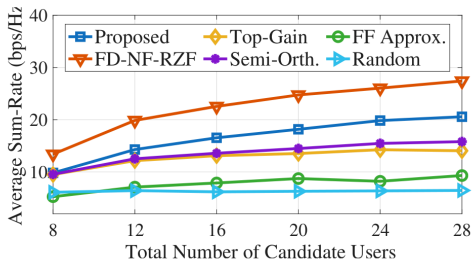


Fig. 3. Average sum-rate versus the total number of candidate users.

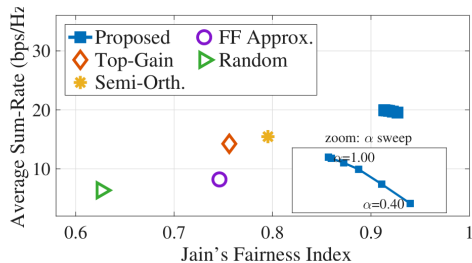


Fig. 4. Throughput–fairness region obtained by sweeping α .

8. References

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Thank You

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